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Resident Social Tolerance–Tourist Bidirectional Feedback Network Flow: *Sustainable Thresholds and Retention-Based Diversion*

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Abstract:

Micro-cultural tourism in historic communities is fundamentally different from that in enclosed scenic areas: the streets and alleys through which visitors walk are also the spaces of residents' daily lives. Therefore, tourist volume is constrained both by the physical carrying capacity of streets and alleys and by residents' social tolerance, and the latter changes dynamically with tourist pressure. Using Zhenjiang's Yinshanmen community as the context, this paper abstracts streets and alleys into a directed network and builds a simplified bidirectional dynamic network flow model based on the Link Transmission Model (LTM). Drawing on the intensity feedback between visitors and residents in tourism social-ecological systems, it introduces a resident social tolerance state and a Sigmoid pressure function, and then makes the socially acceptable flow of each street and alley vary with tolerance, thereby forming a bidirectional feedback system in which tourist flow and resident state interact. This paper proves that, under natural monotonicity conditions, the sustainable external tourist flow rate forms the interval $\left[0, \lambda^{*}\right]$, so a unique sustainable critical flow λ∗ exists. To address the case in which a local street or alley is about to exceed its state-dependent capacity, this paper further constructs a two-branch "capacity-preserving diversion" rule and proves that as long as the total remaining downstream receiving capacity is no less than the flow to be distributed, the rule can always yield a diversion satisfying all downstream capacity constraints; otherwise, total volume control can only be achieved through entrance reservations and delays. With only a few state variables and parameters, the model unifies physical congestion, resident social pressure, and dynamic diversion within a single framework, offering a mathematical explanation for the "small groups, reservations, staggered timing, and diversion" management approach in living historic communities. Numerical examples examine how the critical threshold changes with the distribution of resident sensitivity weights and show how capacity-preserving diversion alleviates overload in narrow alleys. On a 16-node, 20-edge network abstracted from three real themed routes in Yinshanmen, the sustainable critical arrival rate was calculated to be approximately 3.23 persons/min (about 194 persons/h); stress-test comparison experiments show that, while raising the minimum resident tolerance from 0.493 to 0.546, the state-dependent scheme reduces entrance waiting by about 29% compared with static flow limiting, and the maximum pedestrian density across the network is less than 0.5% of jam density, indicating that the carrying-capacity bottleneck in historic communities first appears at the level of resident social tolerance rather than the physical width of streets and alleys.

Keywords:

dynamic network flow; touristflow; social tolerance; state-dependent threshold; capacity-preserving diversion; LTM

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1 Problem Statement

1.1 Basic Concepts

Yinshanmen is not a scenic area built after residents were relocated. It is a “living" historic community where many elderly residents still live: the streets and alleys tourists walk through are also the everyday spaces where residents buy groceries, get around, and rest. So, how many tourists an alley can hold depends not only on how wide it is (its physical carrying capacity) but also on what residents can accept psychologically and in their daily lives (social tolerance).

“resident social tolerance–tourist bidirectional feedback” refers to a closed loop of mutual influence: on the one hand, the more tourists there are, and the more concentrated they are in residential alleys, the lower residents’ tolerance becomes; on the other hand, once tolerance drops, the community tightens the number of tourists allowed into those alleys through staggered timing, small groups, rerouting, and other measures, which in turn affects tourist flow. “network flow” means treating streets and alleys as time-varying pedestrian flows on a directed graph; a “sustainable threshold” means the maximum rate at which visitors can be continuously admitted through the entrance over the long term without letting residents’ tolerance fall below the management floor, denoted λ^*; and “capacity-preserving diversion" means that when a narrow alley approaches its current limit, only minimal intervention is made—the excess tourists are guided to other alleys while preserving visitors’ original natural choices as much as possible.

This paper puts “resident state" into a dynamic network flow model as a variable that changes with tourist pressure and, in turn, limits tourist capacity. It proves that the sustainable tourist flow rate is an interval from 0 to λ^* and calculates λ^*, then gives a set of rules—“do not divert unless necessary, divert as little as needed, and use entrance reservations when total volume exceeds the limit”; finally, it tests the model with data from three real routes in Yinshanmen.

1.2 Background

Yinshanmen is not a tourist attraction built after clearing out its residents. It is a living historic community where people still go about their daily lives. The area contains 29 original streets and alleys, some less than 2 m wide. At the same time, the micro-cultural tourism plan lays out three themed routes and explicitly calls for reservations, staggered timing, and small groups to reduce the disturbance tourists cause to residents’ lives. Tourist management here is therefore subject to two kinds of constraints: first, the physical carrying capacity determined by alley width, length, and opposing pedestrian traffic; second, the social carrying capacity determined by tourist density and the intensity of visitors relative to residents.

Traditional static carrying capacity treats the “permitted number of visitors" as a fixed constant, but the pressure residents actually feel is clearly not constant: when there are few visitors, residents can live normally; when visitors keep concentrating in narrow residential alleys, social pressure rises; and once residents’ tolerance falls, the community responds by staggering timing, shrinking group size, adjusting routes, and taking other steps to reduce the number of visitors entering sensitive alleys. This creates a closed loop: touristflow rate → alley congestion and resident pressure → resident social tolerance → permitted flow and diversion → touristflow rate.

1.3 Research Questions and Approach

This paper focuses on two questions. First, under the bidirectional feedback described above, does a sustainable critical touristflow rate exist, and what conditions determine it? Second, when the total volume has not exceeded the limit but individual streets or alleys are approaching a critical state, how should visitors be diverted to other streets and alleys so that the system never crosses the boundary allowed by the current state?

2 Literature Review

The research most closely related to this paper has two main strands. The first is dynamic network flow modeling for pedestrians and tourists. Bagagiolo et al. use mean-field games to characterize tourist flows through

narrow alleys in historic urban districts, linking tourists' route-choice costs to congestion on the links they occupy [1]. In the broader literature on "time-dependent network flows," the idea that edge travel times vary with load goes back a long way [2]. The Link Transmission Model (LTM) uses sending and receiving functions to efficiently capture, at the network level, the propagation, queuing, and spillback of pedestrian flows [3]. Mai et al.'s PedNStream further develops LTM into a scalable pedestrian network simulation framework, offering ready-to-use components such as width capacity, proportional allocation, and gating control [4]. The physical propagation layer in this paper is precisely a simplification of LTM: it keeps the sending–receiving mechanism but does not introduce finer-grained terms such as stochastic diffusion.

The second strand is tourism carrying capacity, especially social carrying capacity. Tokarchuk et al. argue that social carrying capacity should describe "the upper bound beyond which tourist activity becomes unacceptable to local residents," and they try to estimate this threshold using data such as residents' life satisfaction [5]. Raimondo et al. further stress that carrying capacity is not a fixed number given in advance but a "state-dependent quantity" jointly determined by tourists, residents, facilities, and the environment—one that shifts with the state of the system [6]. This is conceptually consistent with this paper's treatment of critical flow as a quantity that changes with residents' state.

These two strands have so far remained largely independent: pedestrian network flow models mostly consider only physical capacity and congestion; they do not feed residents' social state back into link capacities. Carrying capacity research mostly discusses thresholds at a macro or statistical level and seldom gets down to network flows that can be computed alley by alley and updated hour by hour. For living communities where "life still goes on inside the old houses," there is precisely a need to couple the two.

This paper builds a simple bridge between the two strands: it uses a simplified LTM to describe tourist propagation and two-way spatial competition in narrow alleys, adopts a normalized resident tolerance state to carry forward the idea of social carrying capacity, and then lets the acceptable capacity of each alley vary with that state, creating bidirectional feedback. On this basis, it proves a sustainable flow-rate interval and constructs a preserving diversion. The street-and-alley geometry, population, and route data for the study area come from studies of population and street-and-alley morphology in the Dalongwang Lane Historic District [7–9], as well as from the Yinshanmen Community resource verification report and the Yinshanmen Community micro cultural-tourism planning proposal [10,11]. Because person-by-person measurements of psychosocial parameters are lacking, this paper uses only plausible order-of-magnitude level values and provides a discussion of compatibility and sensitivity.

3 Model Assumptions and Notation

3.1 Model Assumptions

The core sightseeing area of Yinshanmen is abstracted as a directed graph $G=(V, \mathscr{E})$, where V denotes alley entrances, attractions, and touristservice points, and $\mathscr{E}$ denotes directed streets and alleys. A real street or alley that allows two-way traffic is represented by two edges of opposite direction, e and $\bar{e}$.

We consider a finite operating period within a single day and discretize time as $t_n = n\Delta t$, $n = 0, 1, \ldots N_T$. To keep the model from becoming overly complex, we make the following assumptions: the total resident base remains approximately constant over a single day of operation; we study only the macroscopic flow of tourists and do not track individual trajectories; conditions inside each street or alley are treated as uniform within a time step; we retain the spatial competition caused by two-way pedestrian flow but do not add terms such as stochastic diffusion and stochastic lingering from PedNStream [4]; residents' social pressure is determined by tourist exposure intensity and has the feature of "buffering under low pressure and rising rapidly under high pressure."

Let U_e^n and V_e^n denote the cumulative number of tourists who have entered and left edge e by time t_n, respectively. Then the in-alley tourist volume within edge N_e^n, is:

$$N_e^n = U_e^n - V_e^n \tag{1}$$

Where are the alley length, effective width, and effective area, respectively, and the pedestrian density is $k_e^n = \frac{N_e^n}{A_e}$ and A_e the effective area is $A_e = L_e w_e$.

3.2 Model Notation

3.2.1 Network and Geometry

Notation	Meaning	Unit
$G = (V, \mathscr{E})$	Directed street-and-alley network; V is the set of alley entrances/attractions/service points, and $\mathscr{E}$ is the set of directed edges	—
e, $\bar{e}$	A directed street or alley and its opposite-direction edge	—
L_e, w_e	Street/alley length, effective width	meters
$A_e = L_e w_e$	Effective area	square meters
Δt	One time step	minutes
N_T	Total number of time steps in a full day of operation	steps
n_e^f, n_e^ω	Number of time steps taken by free-flow and congestion-wave propagation (positive integer)	steps
$T_e = n_e^f \Delta t$	Free-flow traversal time ()	minutes

3.2.2 Flow and Route Choice

Notation	Meaning	Unit
U_e^n, V_e^n	Cumulative number of people who have entered and left edge e by time t_n	persons
$N_e^n = U_e^n - V_e^n$	Current number of people in the alley; $N_{\bar{e}}^n$ is the number of people in the alley on the opposite-direction edge	persons
$k_e^n = N_e^n / A_e$	Pedestrian density	persons / square meter
k_{jam}	Jam density (taken as 4–6 in this paper)	persons / square meter
c_e	Flow rate per unit width	persons / (meter · minute)
$C_e = c_e w_e$	Physical capacity (flow rate)	persons / minute
S_e^n	Sending amount: number of people that can be sent downstream in this step	persons
R_e^n	LTM physical receiving amount: number of people that can still enter in this step	persons
q_{ij}^n, q_e^n	Number of people transferring from node i to j; total outflow from edge e	persons
$p_{ij}^{0,n}$, p_{ij}^n	Natural (Logit) turning proportion; actual turning proportion	dimensionless
D_j^n	Generalized travel cost, $D_j^n = \frac{T_j^n}{T_{ref}} + \theta \frac{k_j^n}{k_{jam}}$	dimensionless
θ	Weight of the congestion term in the Logit model	dimensionless
T_{ref}	Normalized reference time, taken as 1 minute (may be omitted if time has already been normalized)	minutes
$\mathbf{B} = (B_{\text{er}})$	Edge–path incidence matrix; B_{er}=1 means path r passes through edge e	dimensionless
$b_e(\boldsymbol{\pi}) = \sum_{r \in P} B_{er} \pi_r$	Proportion of total passenger flow that passes through edge e	dimensionless

3.2.3 Social Feedback and Diversion

Notation	Meaning	Unit
R_{res}	Resident base in the study area (treated as constant within a single day rather than modeled as a time-varying resident-population state M as in Ref. [6])	persons
ω_e	Tourist exposure weight of edge e (resident sensitivity)	dimensionless
$z^n = \frac{1}{R_{res}} \sum_{e \in \mathscr{E}} \omega_e N_e^n$	Tourist–resident exposure intensity	dimensionless
$\phi(z) = \frac{1}{1+exp[-\kappa(z-m)]}$	Sigmoid social pressure function	dimensionless
κ, m	Sigmoid steepness; inflection point at which pressure begins to accelerate	dimensionless
a^n	Residents' social tolerance (dynamic, $0 \le a^n \le 1$)	dimensionless
$a^*(z) = \frac{\eta}{\eta+\mu\phi(z)}$	Tolerance equilibrium value for a given z	dimensionless
$a_{\min}$	Acceptable minimum tolerance floor	dimensionless
η, μ	Tolerance recovery rate; pressure depletion rate	min^{-1}
γ_e	Minimum capacity retention ratio	dimensionless
$C_e(a) = C_e[\gamma_e+(1-\gamma_e)a]$	State-dependent acceptable capacity (flow rate)	persons/min
λ	External tourist arrival rate (input variable)	persons/min
$\lambda^*(\boldsymbol{\pi})$	Sustainable critical arrival rate (result quantity)	persons/min
$\boldsymbol{\pi} = (\pi_r)_{r \in P}$	Route proportion vector, satisfying $\sum_{r \in P} \pi_r = 1$	dimensionless
Q^n	Number of people currently awaiting assignment at the junction	persons
$\widehat{R}_j^n = \min\{R_j^n, C_j(a^n)\Delta t\}$	State-dependent actual receiving amount (distinct from R_j^n)	persons
$p_1^n, p_2^n = 1 - p_1^n$	Split proportions in the two directions at a two-way fork	dimensionless
$G^n = Q^n - \widehat{R}_1^n - \widehat{R}_2^n$	Number of people who must be delayed at the entrance (when the total volume exceeds the limit)	persons

4 State-Dependent Dynamic Network Flow Model

4.1 Sending–Receiving Model for Streets and Alleys

This section draws on the Link Transmission Model (LTM) [3,4]. How many tourists an edge can send downstream in one time step depends on the number of people who entered before the free-flow travel time and have not yet left, as well as on the capacity available during that step. Let c_e be the maximum flow rate per unit effective width, and let $C_e = c_e w_e$ be the physical capacity; let n_e^f and $n_e^\omega \in \mathbb{Z}_{>0}$ be the numbers of time steps occupied by free-flow propagation and reverse congestion-wave propagation, respectively (the corresponding physical times are $n_e^f \Delta t$, $n_e^\omega \Delta t$, where the free-flow physical time $n_e^f \Delta t = T_e$ has already been defined in the notation table). Since $t_n - n_e^f \Delta t = t_{n-n_e^f}$, the delayed cumulative quantity can be written uniformly with the discrete subscript $U_e\left(t_n - n_e^f \Delta t\right) = U_e^{n-n_e^f}$. Physically, a tourist walking to the exit at free-flow speed needs n_e^f steps, while information about the remaining upstream space propagates back at congestion-wave speed in n_e^ω steps; therefore, the sending amount uses n_e^f, and the receiving amount uses n_e^ω. The number of tourists that edge e can send downstream at step n (the sending function), S_e^n, is given by Eq. (2):

$$S_e^n = \left[\min\left\{U_e^{n-n_e^f} - V_e^{n-1},\ C_e \Delta t\right\}\right]_+ \tag{2}$$

In Eq. (2), the first term is the boundary sending demand, and the second is the upper bound on sending capacity in this step; $[x]_+ = \max\{x, 0\}$ is used to ensure nonnegativity. This equation follows the standard LTM [3,4] and adopts

PedNStream's width-based capacity: the per-unit-width flow rate C_i in that work is our c_e, so $C_i w_i \Delta t$ is our $C_e \Delta t$; we do not adopt its more refined mixed formulation of actual travel time and density.

Correspondingly, the number of tourists that can still enter edge e in this step (the receiving function), R_e^n, is given by Eq. (3):

$$R_e^n = \left[\min\left\{V_e^{n-n_e^\omega} + k_{\text{jam}} A_e - U_e^{n-1} - N_{\bar{e}}^{n-1},\ C_e \Delta t\right\}\right]_+ \tag{3}$$

The four terms in Eq. (3) means the following: $V_e^{n-n_e^\omega}$ is the cumulative number of departures before the congestion-wave time; $k_{\text{jam}} A_e$ is the maximum number of people that can be accommodated; $-U_e^{n-1}$ subtracts the cumulative number of people who have already entered this edge; and $-N_{\bar{e}}^{n-1}$ further subtracts the number of people in the alley on the opposite-direction edge; k_{jam} is jam density (persons/m², which can be taken as 4–6 persons/m²). Equation (3) retains the two most critical pedestrian features in the PedNStream study [4]: the number of people that can be accommodated is related to the street/alley area A_e; and opposite-direction pedestrian flow $N_{\bar{e}}^n$ shares the same street/alley area with the current direction, so two-way flow in narrow alleys automatically lowers the receivable amount.

4.2 Nodal Transfer and State Update

Let q_{ij}^n be the number of tourists transferred from upstream edge i through a node to downstream edge j at step n (persons/step), and let q_e^n be the total outflow from edge e. It can exceed neither the upstream edge's own sending amount nor the downstream edge's proportionally allocated receiving amount. q_e^n is the number of tourists actually passing through edge e at step n (persons/step); then it exceeds neither its own sending amount nor the receiving amount allocated to the downstream edge according to the diversion proportion. From Eq. (1), $N_e^n = U_e^n - V_e^n$, that is, "the number of people in the alley = cumulative entries − cumulative exits"; in other words, $N_e^n = U_e^n - V_e^n$. Nodal transfer and state update are given by Eq. (4a):

$$\begin{aligned} U_e^{n+1} &= U_e^n + \textstyle\sum_{i:i\to e} q_{ie}^n,\quad V_e^{n+1} = V_e^n + q_e^n \\ N_e^{n+1} &= U_e^{n+1} - V_e^{n+1} = N_e^n + \textstyle\sum_{i:i\to e} q_{ie}^n - q_e^n. \end{aligned} \tag{4a}$$

This means that "the number of people in the alley at the current step = the number of people in the alley at the previous step + inflow during the current step − outflow during the current step," so the number of people on each edge is strictly conserved.

Let the incoming edge of node v be i and the outgoing edge be j, and let p_{ij}^n be the turning proportion (in the absence of intervention, its natural value $p_{ij}^{0,n}$ is given by the Logit model in Section 4.3). The actual transfer quantity is determined by Eq. (4b):

$$q_{ij}^n = \min\left\{p_{ij}^n S_i^n,\ \frac{p_{ij}^n S_i^n}{\sum_{\text{h:}\ \text{h}\to j} p_{\text{hj}}^n S_{\text{h}}^n} R_j^n\right\},\ q_e^n = \sum_{j:\ e\to j} q_{ej}^n. \tag{4b}$$

Here, i: $i\to e$ denotes all upstream edges that flow into edge e, and h: $h\to j$ and j: $e\to j$ are defined similarly. The first term in Eq. (4b), $p_{ij}^n S_i^n$, is the number of people that upstream edge i is willing to send to j; the second term allocates the downstream receiving capacity R_j^n in proportion to each upstream edge's desired demand. Taking the minimum of the two is exactly PedNStream's proportional allocation rule [4]—first, demand is submitted according to each edge's own sending amount, and then all upstream demands are uniformly scaled by their shares in competing for the downstream receiving capacity. In this way, $q_{i\to e}^n$ and q_e^n in Eq. (4a) both have a well-defined source, and $q_{ij}^n \geq 0$ always holds, so neither side's capacity is exceeded.

In Section 8.3.1, the downstream receiving capacity in Eq. (4b) takes different values depending on the control scheme: the no-feedback baseline scheme A uses the physical receiving capacity R_j^n ; the static scheme B uses $\widehat{R}_j^n = \min\{R_j^n, C_j(a_{\min})\Delta t\}$, that is, it fixes tolerance at the management floor; and the proposed scheme C in this paper uses the state-dependent receiving capacity $\widehat{R}_j^n = \min\{R_j^n, C_j(a^n)\Delta t\}$, that is, it replaces R_j^n in Eq. (4b) throughout with $\widehat{R}_j^n$, yielding

$$q_{ij}^n = \min\left\{p_{ij}^n S_i^n, \frac{p_{ij}^n S_i^n}{\sum_{\mathrm{h:h}\to j} p_{\mathrm{h}j}^n S_{\mathrm{h}}^n} \widehat{R}_j^n\right\}$$

and

$$q_e^n = \sum_{j:e\to j} q_{ej}^n$$

When $\sum_{\mathrm{h:h}\to j} p_{\mathrm{h}j}^n S_{\mathrm{h}}^n = 0$, we set $q_{ij}^n = 0$. In this way, residents' social capacity enters as a constraint at every node (including the three-way diversion at hub N_0).

4.3 Natural Route Choice by Tourists

Inspired by the statement in Ref. [1] that "congestion cost rises as the number of people on the branch increases" (Eq. (21) of Ref. [1]), suppose that downstream of the terminal node of edge i, there are several downstream streets and alleys to choose from: let T_j be the free-flow travel time of edge j (a constant, $T_j = n_j^f \Delta t$), and let $O(i)$ be the set of downstream edges available from edge i. In this paper, the normalized density k_j^n/k_{jam} replaces the total number of people on the branch in Ref. [1], and the generalized travel cost of edge j is defined as $D_j^n = \frac{T_j^n}{T_{\mathrm{ref}}} + \theta \frac{k_j^n}{k_{\mathrm{jam}}}$, where T_{ref}=1 min is the reference time, so that the time term and the density term are both dimensionless; $\theta \geq 0$ is the congestion weight (corresponding to the linear coefficient in Ref. [1]; the constant intercept cancels out in the Logit ratio, so it is not kept). Taking utility as the negative of cost, $-D_j^n$, and using the multinomial Logit idea of PedNStream [4], the natural turning proportion $p_{ij}^{0,n}$ with no control is given by Eq. (5):

$$p_{ij}^{0,n} = \frac{\exp(-D_j^n)}{\sum_{j\in O(i)} \exp(-D_{\mathrm{h}}^n)}, \ \sum_{j\in O(i)} p_{ij}^{0,n} = 1. \tag{5}$$

Thus, the shorter the free-flow travel time and the lighter the congestion on a street or alley, the smaller D_j^n is and the more likely it is to be chosen. Without intervention, the actual turning proportion is just the natural proportion, $p_{ij}^n = p_{ij}^{0,n}$; it is not adjusted until the preserving diversion is constructed in Section 7. This section does not solve the full mean-field game of Ref. [1]; it keeps only the core mechanism that congestion states affect tourists' route choice.

5 Social Tolerance of Residents and Bidirectional Feedback

5.1 Tourist–Resident Social Pressure

Let $R_{\mathrm{res}} > 0$ be the resident base in the study area. Unlike Ref. [6], which treats the resident population R as a dynamic state, this paper treats the resident base R_{res} as constant at the single-day operational scale (the continuous form $a(t)$ is used only in the appendix). Because tourists near residential doorways clearly put more pressure on residents than the same number of tourists in an open square, each edge is assigned an exposure weight $\omega_e \geq 0$. Equation (6) defines the community's tourist–resident exposure intensity z^n directly in terms of discrete time steps:

$$z^n = \frac{1}{R_{\mathrm{res}}} \sum_{e\in\mathscr{E}} \omega_e N_e^n \tag{6}$$

Drawing on the Sigmoid social pressure form that Ref. [6] adopts for $\frac{T}{R}$, in Eq. (7), κ denotes the steepness of the Sigmoid, and the social pressure function $\phi(z)$ is defined as:

$$\phi(z) = \frac{1}{1+\exp[-\kappa(z-m)]} \quad \kappa > 0 \tag{7}$$

where m denotes the point at which social pressure begins to rise with a marked acceleration, and κ denotes the steepness of the transition. This distinction matters a great deal: Ref. [6] specifically points out that "physical crowding

caused by the total number of people" and "social pressure caused by the intensity of tourists relative to residents" are not the same thing.

5.2 Residents' Social Tolerance

Let $a(t)$ denote residents' social tolerance: $a = 1$ indicates a favorable state; the smaller a is, the harder it is for residents to accept further increases in tourists. The change in tolerance equals natural recovery when pressure is low minus the loss caused by exposure. Its continuous form is given by Equation (8):

$$\dot{a}(t) = \eta\big(1-a(t)\big)-\mu a(t)\phi\big(z(t)\big) \tag{8}$$

where $\eta, \mu > 0$ (both in min^{-1}) are the recovery rate and loss intensity, respectively. The recovery term $\eta(1-a)$ becomes stronger when a is lower, and the loss term $\mu a\phi(z)$ increases as social pressure ϕ increases. The continuous form is used for property analysis (see Lemma 2 in the appendix for the proof); the numerical iteration uses explicit Euler discretization, that is, all terms on the right-hand side are evaluated directly using the known quantities at step n:

$$a^{n+1} = a^n+\Delta t[\eta(1 - a^n)-\mu a^n\phi(z^n)], a^0 = a_{\text{obs}} \in [0,1] \tag{9}$$

As long as the step size satisfies $\Delta t(\eta+\mu) \le 1$, the iteration is guaranteed to always have $a^n \in [0,1]$ (proof in the appendix). When the exposure level z is fixed, the iteration converges to the unique equilibrium $a^*(z)=\frac{\eta}{\eta+\mu\phi(z)} \in [0,1]$.

6 State-Dependent Acceptable Capacity and Sustainable Critical Threshold

6.1 Residents' State in Turn Constrains Tourist Flow

The fact that "tourists affect residents" alone does not yet constitute two-way feedback. Let $C_e = c_e w_e$ (persons/min) be the physical carrying-capacity constant of edge e. This paper defines the state-dependent acceptable capacity $C_e(a)$ by monotonically interpolating linearly with respect to acceptability a between the minimum capacity $\gamma_e C_e$ and the full capacity C_e:

$$C_e(a) = C_e[\gamma_e+(1-\gamma_e)a], \ 0 < \gamma_e \le 1 \tag{10}$$

Here, γ_e characterizes the sensitivity of streets and alleys to residents' daily life. At the endpoints, $C_e(1)=C_e$ (when acceptability is favorable, social capacity equals physical capacity), and $C_e(0) = \gamma_e C_e$ (the minimum capacity that still retains a proportion γ_e in the worst state); moreover, $\frac{\partial C_e}{\partial a} = C_e(1-\gamma_e) \ge 0$, so $\gamma_e C_e \le C_e(a) \le C_e$ always holds. Open public spaces take a relatively large γ_e (capacity hardly changes with a), whereas narrow residential alleyways take a smaller value.

Two pathways of influence must be distinguished here: tourists' natural route choice $p_{ij}^{0,n}$ (Eq. (5)) depends only on density k_j^n and does not involve a, so it belongs to tourists' spontaneous behavior; the reverse effect of residents' acceptability on tourist flow, by contrast, operates through $C_e(a^n)$ entering the actual admissible intake $\widehat{R}_j^n$ (Eq. (14)) and is then realized via the retention split p_1^n (Eq. (15)) and the critical flow λ^* (Eq. (12)).

Returning to the exposure intensity in Eq. (6), $z^n = \frac{1}{R_{\text{res}}}\sum_{e\in\mathcal{E}}\omega_e N_e^n$: it is not a weighted absolute stock but a per-capita (tourist–resident) exposure intensity obtained by dividing by the resident base R_{res}. It counts only those who have already entered the streets and alleys and does not include those delayed waiting at entrances. When $\omega_e \equiv 1$, $z^n = \frac{1}{R_{\text{res}}}\sum_e N_e^n$ is exactly the network-wide tourist–resident ratio, corresponding precisely to T/R in Ref. [6]; nonuniform ω_e is a spatial-distribution correction that captures the degree of tourist concentration on resident-sensitive streets and alleys. Taken together, these form the negative-feedback loop $N_e^n \to z^n \to \phi(z^n) \to a^n \to C_e(a^n) \to \big(\widehat{R}_j^n, p_1^n\big) \to N_e^{n+1}$.

6.2 Sustainable Critical Flow Rate

Let the community entrance receive tourists at an average rate λ (persons/min), let the set of candidate paths be P, and let the proportions of the paths form a vector $\boldsymbol{\pi} = (\pi_r)_{r\in P}$, satisfying $\sum_{r\in P}\pi_r = 1$. Let the matrix $\mathbf{B} = (B_{er})$ be the edge–path incidence matrix, where $B_{er} = 1$ indicates that path r passes through edge e. Let $b_e(\pi)$ denote the proportion of the total tourist flow that passes through edge e, $b_e(\boldsymbol{\pi})$ $\sum_{r\in P}B_{er}\pi_r$; thus,

when the external arrival rate is λ, the average flow carried by edge e is approximately $\lambda b_e(\pi)$. Once the average rate λ is fixed, the network's long-run average exposure intensity is denoted by $z^*(\lambda, \pi)$. At steady state, setting the tolerance equation $\dot{a} = 0$, i.e. $\eta(1-a) = \mu a \phi\left(z^*\right)$, gives the corresponding socially balanced tolerance $a^*(\lambda, \pi)$, as in Eq. (11):

$$a^*(\lambda,\boldsymbol{\pi}) = \frac{\eta}{\eta+\mu\phi\left(z^*(\lambda,\boldsymbol{\pi})\right)} \tag{11}$$

The derivation of Proposition 1 (see the appendix) uses Little's law $N_e^* = \lambda b_e(\pi) T_e$, whose premise is that "no street or alley is saturated." If for some edge the flow rate satisfies $\lambda b_e(\pi) \le \mathrm{C}_e(a)$ (not exceeding social capacity) but the steady-state inventory $N_e^* = \lambda b_e(\pi) T_e > k_{\mathrm{jam}} \mathrm{A}_e$ (physically full), then the edge is already saturated and Little's law fails early; therefore, the inventory constraint $\lambda b_e(\pi) T_e \le k_{\mathrm{jam}} \mathrm{A}_e$ must be added. This constraint is still monotone in λ, so the interval proof remains valid as is. Given the minimum acceptable tolerance $a_{\min}$, Eq. (12) defines the sustainable critical tourist flow rate $\lambda^*(\pi)$:

$$\lambda^*(\pi) = \sup\{\lambda \ge 0: a^*(\lambda, \pi) \ge a_{\min}, \lambda b_e(\pi) \le \mathrm{C}_e\left(a^*(\lambda, \pi)\right), \lambda b_e(\pi) T_e \le k_{\mathrm{jam}} \mathrm{A}_e, \forall e \in \mathscr{E}\} \tag{12}$$

Here, λ is the given external arrival rate (an input variable), and λ^* is the sustainable upper critical value of λ; Eq. (12) obtains λ^* precisely by taking the supremum over λ. The most fundamental difference between λ^* and a traditional fixed carrying capacity is that λ^* is not a constant given in advance but is jointly determined by the tourist state and the resident state. This idea directly inherits the view in Ref. [6] that "carrying capacity is a state-dependent quantity," but this paper recasts the "tourist inventory equilibrium point" as an "external tourist flow-rate critical value" that better fits street-and-alley operations; its carrying capacity moves with the rest of the system state and is not a fixed upper limit.

Let us now spell out the steady-state exposure intensity more clearly. Again, let the entrance receive tourists at an average rate λ, let the set of candidate paths be P, and let the path proportion vector be $\boldsymbol{\pi} = (\pi_r)_{r \in P}$, satisfying $\sum_{r \in P} \pi_r = 1$; let $\mathbf{B} = (B_{er})$ be the edge–path incidence matrix ($B_{\mathrm{er}} = 1$ means path r passes through edge e, and 0 otherwise). Then the proportion of the total flow that passes through edge e is $b_e(\pi) = \sum_{r \in P} B_{er} \pi_r$, and the average flow rate carried by edge e is $\lambda b_e(\pi)$. In the steady state where no street or alley is saturated, substituting Little's law $N_e^* = \lambda b_e(\pi) T_e$ into Eq. (6) yields the long-run average exposure intensity $z^*(\lambda, \pi)$, as in Eq. (13):

$$z^*(\lambda, \pi) = \frac{\lambda}{R_{\mathrm{res}}} \sum\nolimits_{e \in \mathscr{E}} \omega_e b_e(\pi) T_e \tag{13}$$

7 Preserving Diversion

Knowing the critical flow rate λ^* alone is not enough: the more common situation is that the total volume has not exceeded the limit, but some narrow alley is approaching its critical level because tourists have concentrated there. Consider a fork node with two downstream alleys $j = 1, 2$, and let the number of tourists currently awaiting assignment be Q^n, with natural choice proportions $p_1^{0,n}$ and $p_2^{0,n}$. The actual receivable amount is constrained by both physical congestion and the residents' state; we take the smaller of the two, as in Eq. (14):

$$\widehat{R}_j^n = \min\{R_j^n, C_j(a^n)\,\Delta t\}, \forall j \in \mathrm{O}(i) \tag{14}$$

The two capacity constraints are equivalent to the feasible interval $1 - \frac{\widehat{R}_2^n}{Q^n} \le p_1^n \le \frac{\widehat{R}_1^n}{Q^n}$. Preserving diversion pushes the natural proportion back into this interval only when it falls outside it, as in Eq. (15):

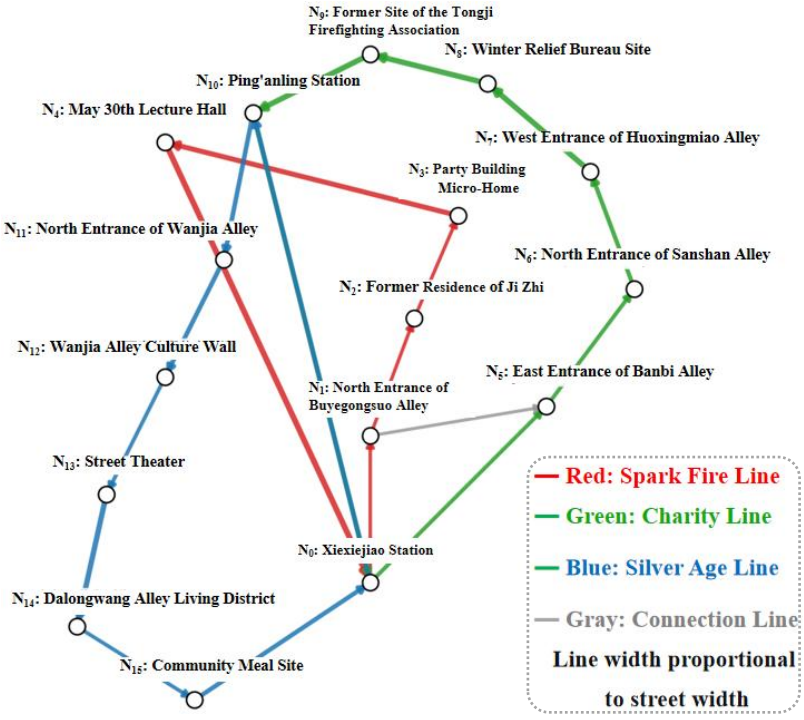
$$p_1^n = clip\left(p_1^{0,n}, 1 - \frac{\widehat{R}_2^n}{Q^n}, \frac{\widehat{R}_1^n}{Q^n}\right) = \min\left\{\max\left\{\left\{p_1^{0,n}, 1 - \frac{\widehat{R}_2^n}{Q^n}\right\}, \frac{\widehat{R}_1^n}{Q^n}\right\}\right\}, p_2^n = 1 - p_1^n \tag{15}$$

Equation (15) uses the interval projection (clip) operator; when $Q^n = 0, p_1^n = p_1^{0,n}(clip\,(x, l, u)) = \min\left\{\max\{x, l\}, u\right\}$; Eq. (15) is not activated when $Q^n = 0$. It preserves tourists' natural choices as much as possible and intervenes only minimally, rather than re-optimizing everyone's behavior. If total demand exceeds the sum of the two downstream capacities, no diversion can satisfy both constraints simultaneously; in that case, the only option is to delay $G^n = Q^n - \widehat{R}_1^n - \widehat{R}_2^n$ people at the entrance (see Proposition 2 in the appendix).

8 Numerical Experiment

8.1 Network Construction and Parameter Settings

This section constructs a computational network using Yinshanmen's three themed routes—the Red Spark Route, the Charity Culture Route, and the Silver-Age Life Route—as prototypes. All three routes use the "Xiexiejiao" Rest Station (located at the Community Party-Mass Service Center, No. 8-1 Dalongwang Lane, and serving as the unified starting and ending point for the entire community) as their common hub. While retaining each route's key interpretive nodes and closed-loop routing, lane entrances, cultural-relic interpretive points, and service points are abstracted into 16 nodes（N_0, N_1, ⋯ N_{15}), and the touring lane segments between adjacent nodes are abstracted as directed edges. Actual two-way lanes are represented by two oppositely directed edges, following the assumption in Section 3.1. An additional connector edge links the Red Spark Route and the Charity Culture Route. The result is a network $G = (V, \mathscr{E})$ with 16 nodes and 20 directed edges, as shown in Figure 1. It should be noted that although the jurisdiction actually contains 29 original lanes, this network does not reproduce them one by one. Instead, using Dalongwang Lane as the basis, and without changing the three planned circulation routes or their flow-distribution relationships, it reasonably aggregates the lane segments relatedd to touristflow: lane lengths are measured to scale from the planned route map, and effective widths are assigned according to field-based classification (main lanes: 3.0–4.5 m; general lanes: 2.0–3.0 m; branch lanes: 1.0–2.0 m).



Node	Corresponding Real-World Location
N_0	Xiexiejiao Station
N_1	North Entrance of Buyegongsuo Alley
N_2	Former Residence of Ji Zhi
N_3	Party Building Micro-Home
N_4	May 30th Lecture Hall
N_5	East Entrance of Banbi Alley
N_6	North Entrance of Sanshan Alley
N_7	West Entrance of Huoxingmiao Alley
N_8	Winter Relief Bureau Site
N_9	Former Site of the Tongji Firefighting Association
N_{10}	Ping'anling Station
N_{11}	North Entrance of Wanjia Alley
N_{12}	Wanjia Alley Culture Wall
N_{13}	Street Theater
N_{14}	Dalongwang Alley Living District
N_{15}	Community Meal Site

Figure 1 Network topology of the three themed routes in Yinshanmen Community

As shown in Table 1, the parameters are divided into three tiers by credibility: geometric and demographic parameters (L_e, w_e, R_{res}) come from measurements taken from route maps, street-and-alley classification, and neighborhood population literature [7–9]; pedestrian traffic parameters (v_f, v_w, c^*, k_{jam}) are taken from commonly used ranges in pedestrian traffic engineering; and social feedback parameters ($\kappa, m, \eta, \mu, a_{\min}$) currently lack person-by-person empirical measurement, so only level values that are plausible in order of magnitude and that satisfy baseline compatibility and monotonicity are used. In particular, the social pressure steepness κ cannot be chosen arbitrarily: when there are no

tourists, $\phi(0) = 1/(1+e^{km})$, corresponding to the steady-state tolerance $a^*(0) = \eta/[\eta + \mu\phi(0)]$. If one takes the relatively gentle steepness commonly found in the literature, $\kappa = 5$, then $a^*(0) = 0.438 < a_{\min} = 0.6$, which means that "even with no tourists at all, the residents' baseline tolerance is below the management floor"; the feasible set is empty and the critical value degenerates to 0. This contradicts the fact that, at Yinshanmen, "residents were in good condition before opening, and the questionnaire showed that most elderly residents were willing to tell stories about the old streets." When $\kappa = 14.5$, $a^*(0) = 0.710 > a_{\min}$, and the feasible set is nonempty. This value is used only to determine the shape setting of the Sigmoid's "slow rise–acceleration" profile; the critical value is then solved independently from Eq. (12).

Table 1. Parameter meanings and values

Symbol	Meaning	Value	Basis for value (not questionnaire-fitted)
Δt	Discrete time step	0.5 min	Taken as the order of magnitude of the free-flow travel time through the shortest lane segment
T	Operational calculation duration	120 min	Corresponds to a half-day concentrated sightseeing window
v_f	Free-flow walking speed	0.9 m/s (54 m/min)	Age-friendly slow-touring pace, taken as the midpoint of the commonly used slow-walking range of 0.8–1.0 m/s
$\lvert v_w \rvert$	Congestion wave backward propagation speed	1.8 m/s	As is conventional in the triangular fundamental diagram, vw ≈ −2vf; same as Ref. [4]
c^*	Passage rate per unit effective width	60 persons/(min·m)	Converted from vf and the upper comfort density limit; same order of magnitude as Ref. [4]
k_{jam}	Jam density	5.4 persons/m²	Midpoint of the commonly used pedestrian jam-density range of 4–6 persons/m²
θ	Logit dispersion	1	Dimensionless baseline value; the cost in Eq. (5) is already dimensionless
a^0	Initial tolerance	0.95	Before opening, no visitors are present and residents are in good condition
R_{res}	Resident base of the study area	750 persons	About 350 households and 750 people in the Dalongwang Lane neighborhood
κ	Steepness of the social-pressure sigmoid	14.5	Calibrated by the compatibility condition that the zero-touristbaseline tolerance must not fall below the bottom line
m	Pressure acceleration inflection point (exposure intensity)	0.15	Inflection-point level for low-pressure buffering and high-pressure acceleration
η	Natural recovery rate of tolerance	0.02 min⁻¹	Recovery on a time scale of about several tens of minutes
μ	Exposure depletion rate	0.08 min⁻¹	Depletion is faster than recovery, consistent with cumulative stress characteristics
$a_{\min}$	Minimum acceptable tolerance	0.6	Management bottom line (corresponding to "acceptable to most residents")
$\boldsymbol{\pi}$	Admission proportions of the three routes	$(1/3, 1/3, 1/3)^{\mathrm{T}}$	Baseline scenario of balanced reception across the three routes

8.2 Calculation of the Sustainable Critical Arrival Rate

Before running the dynamic simulations, we first use Proposition 1 to obtain the analytical critical value, which serves as a benchmark for the subsequent stress-testing experiments. Fix the path proportions for balanced reception at $\boldsymbol{\pi} = (1/3, 1/3, 1/3)^{\mathrm{T}}$, and let $\mathbf{B}$ be the edge–path incidence matrix (with entry 1 if the path traverses that edge and 0 otherwise). Then the vector of external arrival proportions borne by each edge is $b = \mathbf{B}\boldsymbol{\pi}$. By Little's law, in steady state, the average number of people in the alley on edge e is $\lambda b_e T_e$, where $T_e = L_e / v_f$ is the free-flow traversal time. The long-run average exposure intensity over the entire network is

$$z^*(\lambda) = \frac{1}{R_{\text{res}}} \sum_e \omega_e (\lambda b_e T_e) = \frac{\lambda H}{R_{\text{res}}},$$

Where $H := \sum_{e \in \mathcal{E}} \omega_e b_e(\pi) T_e$.

Substituting the network parameters (the free-flow walking times of the three themed routes are approximately 11.5, 13.3, and 11.7 min, respectively), we obtain $H = 9.0679$ min. Setting the steady-state tolerance equation to equality:

The steady-state tolerance and social pressure are, respectively, $a^*(z) = \frac{\eta}{\eta+\mu\phi(z)}$, $\phi(z) = \frac{1}{1+\exp[-\kappa(z-m)]}$.

Set the management floor at $a_{\min} = 0.6$. Let $a^*(\lambda^*) = a_{\min}$. First, solve backward for the critical pressure:

$$\varphi^* = \frac{\eta(1-a_{\min})}{\mu a_{\min}} = \frac{0.02\times0.4}{0.08\times0.6} = 0.1667.$$

Then, using the inverse Sigmoid, we obtain the critical exposure intensity:

$$z^* = m+\frac{1}{\kappa}\ln\frac{\varphi^*}{1-\varphi^*} = 0.15+\frac{1}{14.5}\ln\frac{0.1667}{0.8333} = 0.0390$$

Therefore, the sustainable critical external arrival rate is

$$\lambda^* = \frac{R_{\text{res}}z^*}{H} = \frac{750\times0.0390}{9.0679} = 3.23 \text{ persons/min} \approx 194 \text{ persons/h}.$$

As a cross-check, we directly perform a bisection search on the three types of constraints in Eq. (12) (social tolerance, state-dependent capacity, and physical capacity). This gives $\lambda^* = 3.226013$ persons/min, which differs from the closed-form solution above, 3.226018 persons/min, by only 4.7×10^{-6} persons/min. At $\lambda = \lambda^*$, checking each edge one by one shows that the physical passage capacities are nowhere near being reached. Therefore, the constraint that actually binds is the social tolerance condition. Figure 2 shows how $a^*(\lambda)$ and $z^*(\lambda)$ vary with λ: in the left panel, the intersection of the decreasing a^* and the horizontal threshold $a_{\min}$ is exactly λ^*; the right panel verifies that z^* is linear in λ.

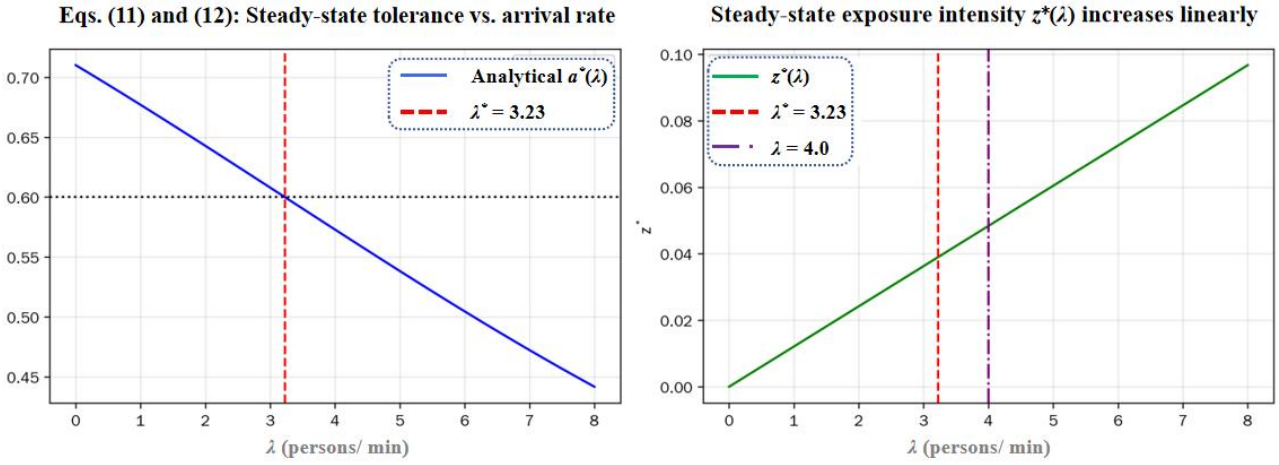


Figure 2 Analytical curves of the critical threshold

It should be emphasized that $\lambda^* = 194$ persons/h is an early-warning upper bound under a scenario of concentrated tourist arrivals. Yinshanmen micro-cultural tourism normally operates with small booked groups (10–50 people per group) and staggered reception, and its planned daily reception volume is on the order of hundreds, far below this upper bound [10,11]. This paper sets $\lambda = 4.0$ persons/min (about 1.24 times λ^*) in order to stress-test the unfavorable scenario in which multiple small groups arrive together on the same morning, or the system must absorb the transient spillover from the adjacent Xijindu Historical and Cultural District (which received nearly 300,000 visits over the five-day May Day holiday in 2026).

8.3 Comparative Experimental Design and Evaluation Metrics

8.3.1 Comparative Experimental Design

The numerical experiments compare how different control schemes affect street-and-alley density, resident tolerance, entrance waiting, and in-network dwell time. Whether state-dependent control improves these metrics, and the waiting and detour costs required for such improvement, are judged from the computational results. As shown in Table 2, three schemes are compared in total, and all three must use exactly the same external arrival demand, network, initial state, time step, and parameters.

Table 2 Comparative Experimental Design

Scheme	Control method	Purpose
A (Baseline)	Subject only to physical capacity constraints (tolerance does not enter the capacity constraints, i.e.$a \equiv 1$; no entrance flow limiting, and all arrivals enter; both the entrance and internal forks follow the natural Logit route choice in Eq. (5) and the proportional allocation in Eq. (4b)	Serves as the no-feedback baseline
B (Static carrying capacity)	Social capacity is fixed at $C_e(a_{\min})$, with the management floor $a_{\min} = 0.6$; the entrance is subject to a fixed flow limit at the critical rate $\lambda^* = 3.226$persons/min persons/min, and those exceeding it queue at the entrance; route choice is the same as A	Tests the traditional static idea of a "fixed carrying capacity"
C (This paper's method)	Social capacity is dynamically updated according to the state-dependent capacity formula $C_e(a^n)$ as the network-wide tolerance a^n changes; the entrance is flow-limited by the state-dependent feedback law ($\widehat{R}_j^n = \min\{R_j^n, C_j(a^n)\Delta t\}$), and those exceeding it queue at the entrance; the two forks are diverted according to the preserving diversion formula (clip rule)	Tests this paper's "state-dependent capacity + preserving diversion"

8.3.2 Evaluation Metrics

Let q_e^n be the number of tourists who actually pass through edge e at step n (persons/step). q_e^n comes from the nodal transfer equations (4a) and (4b), and is constrained by the sending function (2) and the receiving function (3). The numerical experiments report the following five metrics.

(1) The maximum pedestrian density $k_{\max}$ within the same physical street or alley (persons/m²), as in Eq. (16):

$$k_{\max} = \max_{0 \le n < N_T} \max_{e \in \mathscr{E}} \frac{N_e^n + N_{\bar{e}}^n}{A_e} \tag{16}$$

(2) The time T_{below} during which resident tolerance falls below the limit (min), as in Eq. (17):

$$T_{\text{below}} = \Delta t \sum_{n=0}^{N_T-1} \mathbf{1}\left\{\min_e a^n < a_{\min}\right\} \tag{17}$$

(3) The lowest resident tolerance a_{low} (dimensionless), as in Eq. (18):

$$a_{\text{low}} = \min_{0 \le n < N_T} \min_{e \in \mathscr{E}} a^n \tag{18}$$

(4) The cumulative entrance waiting time W_{delay} (person·min), as in Eq. (19):

$$W_{\text{delay}} = \Delta t \sum_{n=0}^{N_T-1} G^n \tag{19}$$

where G^n is the number of tourists who cannot enter at step n and must wait in queue at the entrance.

(5) The average network dwell time of tourists $\overline{T}$ (min), as in Eq. (20):

$$\overline{T}_{\text{net}} = \frac{\Delta t \sum_{n=0}^{N_T-1} N^{\text{tot},n}}{M_{out}}, N^{\text{tot},n} = \sum_{e \in \mathscr{E}} N_e^n, M_{\text{out}} > 0 \tag{20}$$

Here, if the network starts empty, is fully cleared by the end, and each tourist enters and leaves only once during the trip, let the total number of completed tourists be $M_{out} > 0$. Eq. (20) is built on the discrete-time approximation $T_j^n = T_e^f$ and is a discrete-time Little's-law approximation. If tourists are still in the network or queuing at the end of the window, M_{out} is taken as the number who have left within the window; this metric is therefore used for side-by-side comparison of the three schemes over the same window.

To test how well bidirectional feedback and preserving diversion actually work, under the same network, same external demand, same initial state (empty network, $a^n = 0.95$), and same time step, we flip only the “control switch" and compare three schemes: baseline A with no state feedback, static B with entrance flow limiting at a fixed rate, and scheme C with state-dependent capacity (Eq. ((10)) combined with preserving diversion (Eq. (15)). External arrivals follow a trapezoidal profile λ^n: they rise linearly from 0 to $\lambda = 4.0$ persons/min over the first 15 min, stay constant for the middle 90 min, and fall linearly back to 0 over the final 15 min. The total demand $\sum_n \lambda^n \Delta t$ is 420 persons for all three schemes. Thus, if scheme C performs better, the improvement can only be attributed to state-dependent capacity and preserving diversion themselves, not to its receiving fewer tourists.

8.4 Results Analysis

Figure 3 shows the temporal evolution of the core variables under the three schemes. In Scheme A, the network-wide number of people in the lanes peaked at 78.9 persons at $t \approx 46$ min and then remained high; residents' tolerance fell below the bottom line starting at $t \approx 47.5$ min and reached the deepest trough ($a_{\text{low}} = 0.493$), and there was no entrance queuing throughout. Scheme B used the strongest entrance restriction to bring the peak number of people in the lanes down to 63.7 persons and had the latest drop below the bottom line ($t = 57.5$ min), but entrance queues began accumulating early in the pressure phase and peaked at 71.9 persons. In Scheme C, the number of people in the lanes reached 75.7 persons at $t \approx 39$ min and then actively declined in accordance with the state-dependent capacity; its tolerance trough was the highest (0.546), and queuing started later and peaked lower (68.4 persons). The peak exposure intensities under the three schemes were 0.0806, 0.0650, and 0.0768, respectively.

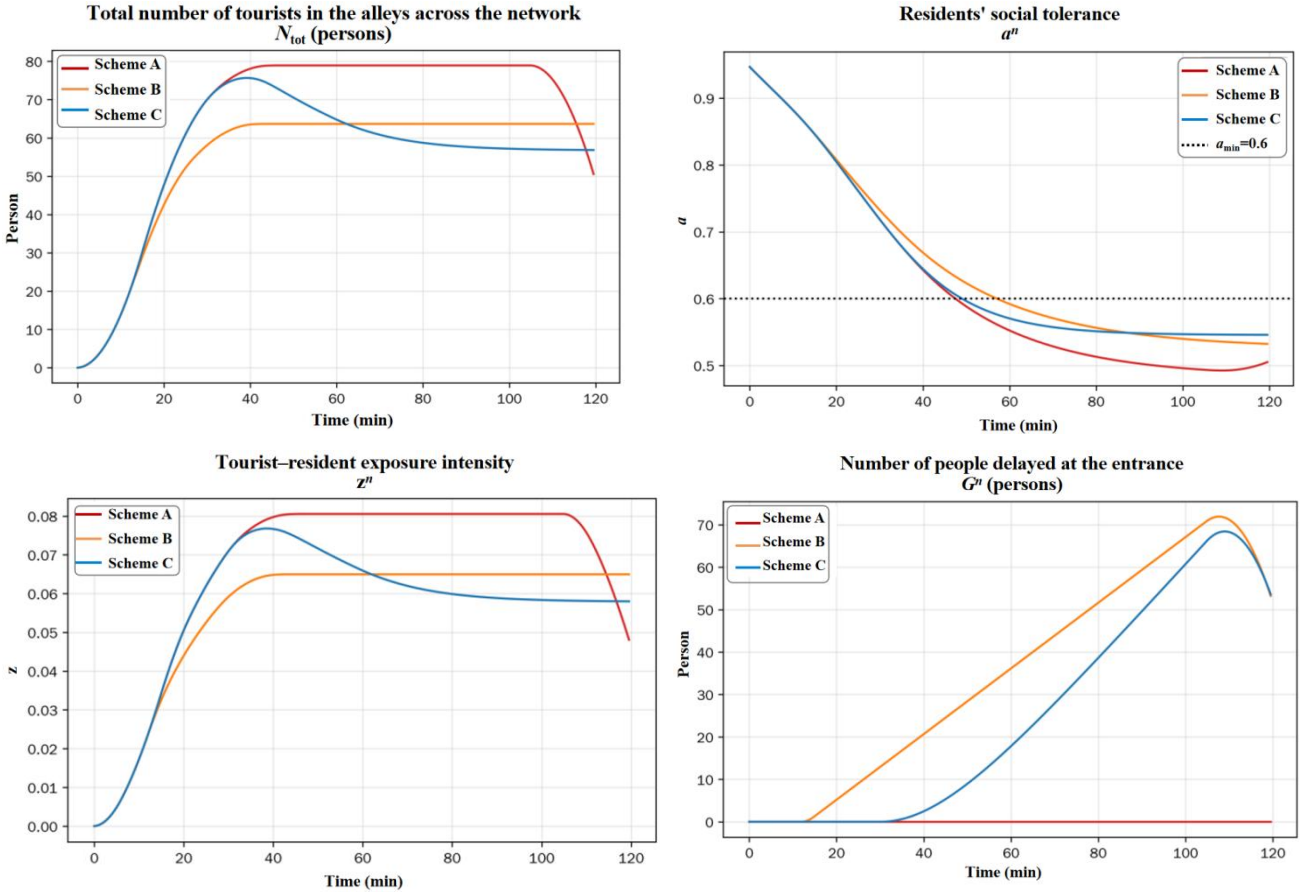


Figure 3 Dynamic comparison time series for the three schemes

Figure 4 presents, as a heatmap, the maximum pedestrian density on the 20 edges over the entire process. The densest edge is the connector edge at the community meal-assistance point (e_{17}), with values of 0.026, 0.021, and 0.026 persons/m² for the three schemes, respectively. The connector edge at Weitongxing Lane (e_{19}), the narrowest edge and the one designated as sensitive, has a density of only about 0.006 persons/m². The maximum density in the entire network is only 0.39%–0.48% of the jam density $k_{\text{jam}} = 5.4$ persons/m^2, and it is also far below the lower bound for pedestrian comfort density. This shows that under this stress scenario, the physical circulation space is far from saturated, and the first boundary to be reached is residents' social tolerance—that is, the binding "bottleneck" lies at the social layer, not at the street-geometry layer.

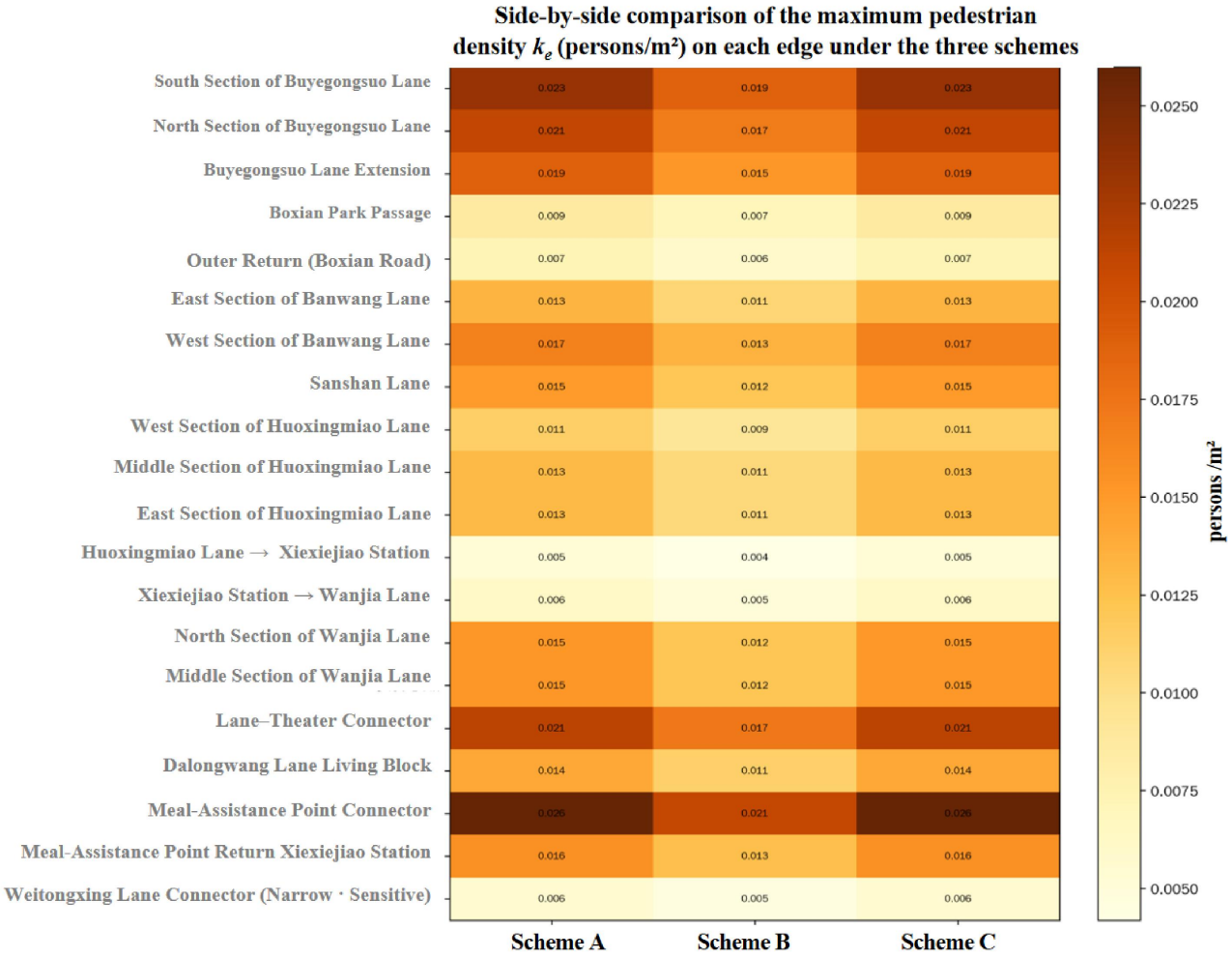


Figure 4 Heatmap of the maximum pedestrian density on each edge over the entire processs under the three schemes

The values of the five metrics above under the three schemes are summarized in Table 3 and compared visually in the bar chart in Figure 5:

Table 3 Comparison of the five metrics under the three schemes (λ = 4.0persons⁄min; total demand = 420 persons).

Metric	Scheme A No feedback	Scheme B Static flow limiting	Scheme C State-dependent (this paper)	Best performer
Maximum density k_{max}	0.026	0.021	0.0259	B
Time below the tolerance limit T_{below} (min)	72.5	62.5	70.5	B
Minimum tolerance a_{low}	0.4925	0.5324	0.5459	C
Cumulative entrance waiting W_{delay} (person·min)	0	4236.3	3017.6	A (C is smaller among the controlled schemes)
Average network dwell time $\bar{T}_{net}$ (min)	21.24	21.58	21.35	A (B and C are nearly the same)

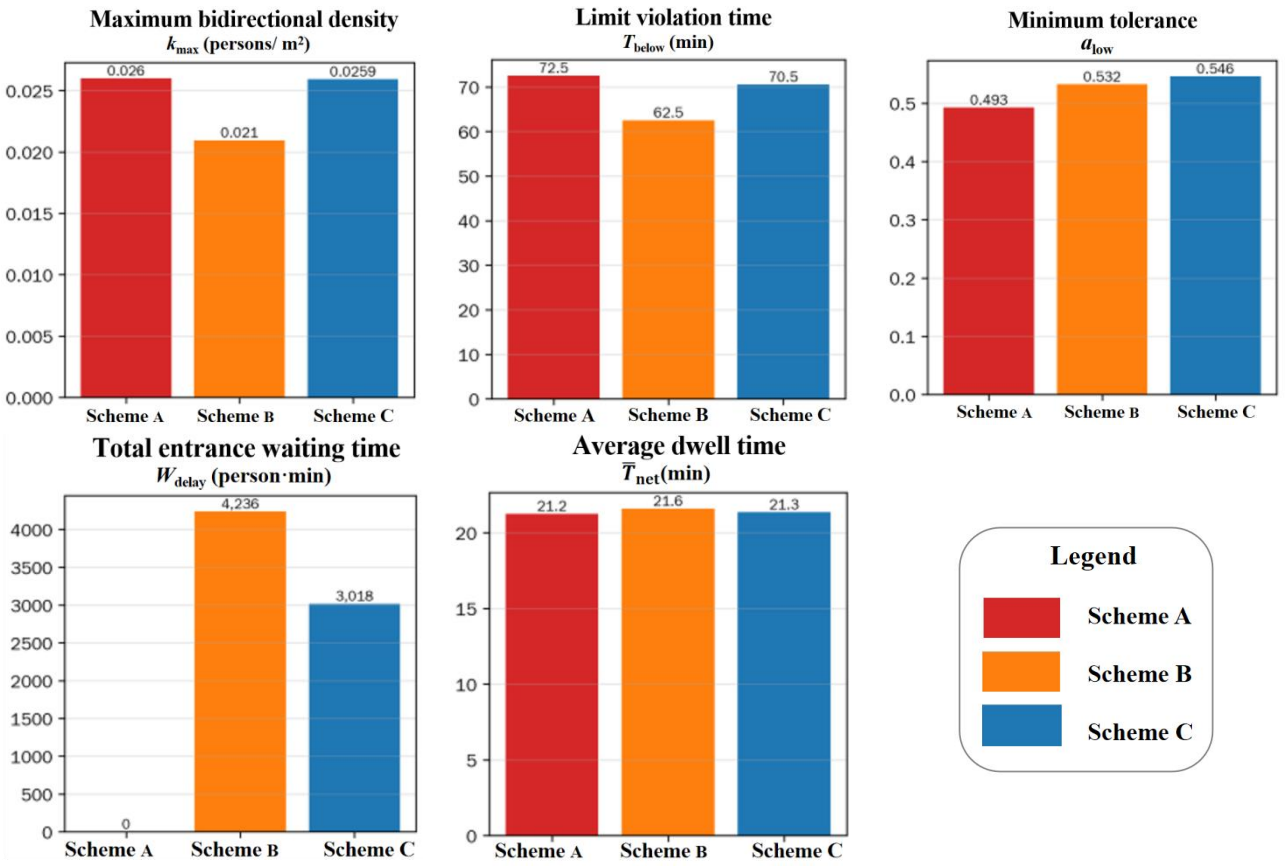


Figure 5 Comparative evaluation of no-feedback, static, and state-dependent control schemes under high-demand stress (λ = 4.0 persons/min)

The most valuable phenomenon in the numerical results is not that "C is best on every metric," but rather a clear set of trade-offs: Scheme C raises the minimum resident tolerance from 0.493 under A to 0.546 (also higher than 0.532 under B), reduces cumulative entrance waiting by about 29% relative to static scheme B (3018 vs. 4236 person·min), and has a slightly shorter average dwell time (21.35 vs. 21.58 min). The cost is that its maximum pedestrian density is slightly higher than B's (0.0259 vs. 0.0210 persons/m²), and the duration for which tolerance falls below the floor is slightly longer than B's (70.5 vs. 62.5 min). In other words, the state-dependent scheme trades controllable density and exceedance duration for a substantial increase in the resident tolerance floor and a marked reduction in tourist waiting; at extreme peaks, entrance delays are still needed. This is a reasonable trade-off, and it is precisely where the value of state-dependent feedback and preserving diversion lies.

To directly test the preserving diversion rule (Eq. (15)), we set up a separate "narrow branch alley–wide connector alley" fork unit. When the tolerance of the narrow alley drops to about 0.35, its state-dependent capacity shrinks to about 6.76 persons/min, while the physical receiving capacities of the two alleys are $C_n = 11.7$ and $C_w = 33.6$ persons/min, respectively. Tourists naturally prefer the narrow alley as a shortcut, with a natural proportion $p_1^n = 0.97$. Figure 6 clearly shows what "preserving" means: when tolerance is sufficient ($a \approx 1$), the state-dependent capacity of the narrow alley is $C_e(a) = C_n[\gamma_n + (1-\gamma_n)a] = 11.7$ persons/min, which is greater than the natural flow $p_0 Q \approx 8.9$ persons/min; hence $p_1^n = p_0$, Eq. (15) does not intervene at all, and 8.92 persons/min enter the narrow alley while 0.28 persons/min enter the wide alley. When residents along the narrow alley are under pressure ($a = 0.35$, sensitivity coefficient $\gamma = 0.35$), the state-dependent feasible interval is compressed to [0, 0.734]. Eq. (15) projects only the out-of-bounds 0.97 onto the upper bound $p_1^{\max} = 6.769.2 \approx 0.73$; inflow to the narrow alley drops to 6.76 persons/min, and 2.44 persons/min is diverted to the wide alley. Once the pressure is relieved, the proportion returns on its own to 0.97 (Figure 6). This shows that Eq. (15) strictly satisfies the preserving design principle: "do not intervene when the natural proportion is feasible; make only the minimal correction when it goes out of bounds."

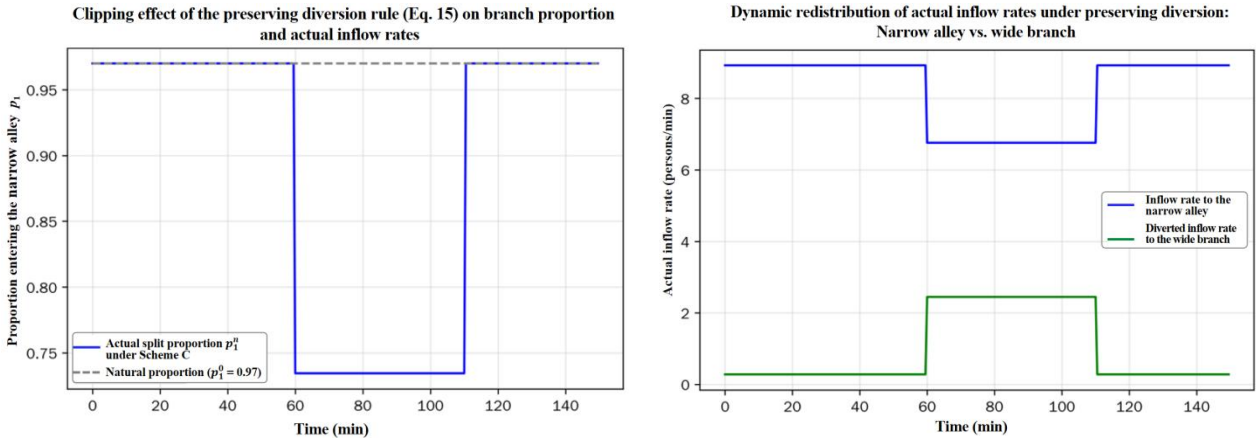


Figure 6 Mechanism of the preserving diversion (Eq. 15): State-dependent split ratio adjustment and actual flow redistribution

9 Conclusions

This paper addresses the interaction between tourist flows and residents' living space in living historic communities by developing a state-dependent dynamic network flow model under bidirectional feedback between residents' social tolerance and tourist flows. A simplified LTM is used to describe tourist propagation and two-way spatial competition in narrow alleys. A Sigmoid tourist–resident pressure function and a resident social tolerance state a^n are introduced so that tourist flows alter the social state. The acceptable capacity of streets and alleys, $C_e(a^n)$, is then allowed to vary dynamically with a^n, so that the resident state in turn constrains the tourist network. This forms a closed-loop feedback.

Under natural monotonicity conditions, this paper proves that sustainable external arrival rates form the interval $\left[0, \lambda^* \right]$, so a unique sustainable critical flow rate λ^* exists. The preserving diversion rule constructed in Eq. (15) further shows that when the total remaining downstream receiving capacity is no less than the flow to be assigned, there must exist a feasible diversion that satisfies the state-dependent capacity constraints of each branch. When total demand exceeds the sum of downstream capacities, however, internal diversion alone is ineffective, and total volume control must be completed through reservations and entrance delays.

For the three-route Yinshanmen case study, we obtain $\lambda^* \approx 3.23$ persons/ min (about 194 persons/h), and show that the state-dependent scheme reduces cumulative entrance waiting by about 29% relative to static flow limiting, while raising the trough of resident tolerance to 0.546 (higher than 0.532 under the static scheme and 0.493 under the no-feedback baseline). Therefore, for historic communities where residents still genuinely live, the reasonable goal of tourist management is not to pursue a fixed notion of the "maximum number of tourists," but to update the acceptable boundary in real time according to resident and network states, and to keep tourist flow within that boundary as much as possible.

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Appendix A: Proofs of Theoretical Propositions and Lemmas

Proposition 1 Interval property of sustainable flow

In the steady state where no street or alley is saturated, the tourist inventory on edge e is $N_e^* = \lambda b_e(\boldsymbol{\pi}) T_e$. By Little's law, substituting $N_e^* = \lambda b_e(\boldsymbol{\pi}) T_e$ (steady-state inventory = arrival rate × traversal time) into Eq. (6) gives $z^*(\lambda, \boldsymbol{\pi}) = \frac{\lambda}{R_{\text{res}}} \sum_{e \in \mathcal{E}} \omega_e b_e(\boldsymbol{\pi}) T_e$.The right-hand side of this expression is a linear function of λ with a positive slope; we assume below that $\sum_{e \in \mathcal{E}} \omega_e b_e(\boldsymbol{\pi}) T_e > 0$, i.e., that there is at least one edge used by a path and sensitive to residents. Under this assumption, $z^*(\lambda, \boldsymbol{\pi})$ is continuous and strictly increasing in λ.

Proof: If $\lambda_1 < \lambda_2$, monotonicity gives $z^*(\lambda_1, \pi) \le z^*(\lambda_2, \boldsymbol{\pi})$. Since ϕ is increasing, the ODE in Eq. (8) gives $a^*(\lambda_1, \boldsymbol{\pi}) \ge a^*(\lambda_2, \boldsymbol{\pi})$. Then, from the state-dependent capacity formula in Eq. (10), $C_e\left(a^*(\lambda_1, \pi)\right) \ge C_e\left(a^*(\lambda_2, \boldsymbol{\pi})\right)$. Therefore, if the larger flow λ_2 satisfies all constraints in Eq. (12), then any smaller flow λ_1 must also satisfy them. Hence, the set of sustainable flows must be an interval starting at 0, provided that $a^*(0) \ge a_{\min}$, i.e., $\frac{\eta}{\eta + \mu\phi(0)} \ge a_{\min}$. On the other hand, each edge has finite physical capacity, so the interval must have a finite upper bound; therefore, a unique boundary value λ^* exists. If $\frac{\eta}{\eta + \mu\phi(0)} < a_{\min}$, then the feasible set is empty, and we set $\lambda^* = 0$, meaning that no sustainable positive flow exists when the minimum tolerance floor is incompatible with the recovery/depletion parameters. This completes the proof. This conclusion is consistent with the mathematical logic in Ref. [6], which obtains a unique state-dependent capacity by means of "the net tourist growth rate strictly decreasing with tourist volume," except that this paper replaces that condition with a network capacity constraint.

Lemma 1 (Positive Scaling Invariance of the Logistic Function)

Let $\phi_{\kappa,m}(x) = \frac{1}{1+\exp[-\kappa(x-m)]}$, where $\kappa > 0$, $m \in \mathbb{R}$. Then for any $c > 0$, $\phi_{\kappa, m}(cx) = \phi_{\kappa c, m/c}(x)$.

Proof: $\phi_{\kappa, \mathrm{m}}(cx) = \frac{1}{1+\exp[-\kappa(cx-m)]} = \frac{1}{1+\exp[-(\kappa c)(x-m/c)]} = \phi_{\kappa c, m/c}(x)$.

The lemma shows that multiplying the argument of the Sigmoid by a positive constant c is equivalent to multiplying the steepness κ by c and dividing the inflection point m by c. The function family itself is unchanged; only the parameters are rescaled. When the weights ω_e are fixed, $z(t)$ is a weighted exposure intensity, a form widely used in the macro socio-ecological literature (e.g., Raimondo et al.) [6]. In other words, the z in this paper differs from the $\frac{T}{R}$ in Ref. [6] only by a positive factor, and in steady state (Little's law), that factor is exactly constant, so the two differ only in their parameter values.

This completes the proof.

Proposition 2 Feasibility of Preserving Diversion

If $Q^n \le \widehat{R}_1^n + \widehat{R}_2^n$, then there exists a feasible diversion proportion $p_1^n \in [0,1]$ satisfying $p_1^n Q^n \le \widehat{R}_1^n$ and $p_2^n Q^n \le \widehat{R}_2^n$.

Proof: The interval allowed by Eq. (15) is nonempty if and only if $1 - \frac{\widehat{R}_2^n}{Q^n} \le \frac{\widehat{R}_1^n}{Q^n}$, which, upon rearrangement, is exactly $Q^n \le \widehat{R}_1^n + \widehat{R}_2^n$. Therefore, under the condition $Q^n \le \widehat{R}_1^n + \widehat{R}_2^n$, a feasible p_1^n must exist. Since $p_1^n \le \frac{\widehat{R}_1^n}{Q^n}$, we have $p_1^n Q^n \le \widehat{R}_1^n$. Also, since $p_1^n \ge 1 - \frac{\widehat{R}_2^n}{Q^n}$, we have $p_2^n Q^n = (1 - p_1^n) Q^n \le \widehat{R}_2^n$. Thus neither direction exceeds its own state-dependent capacity. This completes the proof.

Conversely, if $Q^n > \widehat{R}_1^n + \widehat{R}_2^n$, then no redistribution of proportions can solve the problem. In this case, $G^n = Q^n - \widehat{R}_1^n - \widehat{R}_2^n$ tourists must be delayed at the entrance. When $Q^n = 0$, no diversion is needed. This yields an important conclusion: diversion can address only "uneven spatial distribution"; it cannot address "total volume exceeding the carrying capacity of the entire network." Therefore, the "internal flow diversion" at Yinshanmen and the "reservation system, small-group system, and staggered scheduling" are not measures of the same kind: the former changes local diversion proportions (Eq. (15)), whereas the latter changes the entrance flow rate λ. This

is also consistent with the idea in PedNStream of using gating to limit inflow and alter capacity and route choice [4].

Lemma 2 (Invariant Interval of Tolerance)

If $a(0) \in [0,1]$, then the solution of model (8) always satisfies $0 \leq a(t) \leq 1$. Let $a(t) \in [0,1]$ denote residents' social tolerance: $a = 1$ indicates a good state, and the smaller a is, the harder it is to accept further increases in tourists. Equation (8) is a continuous ODE that defines tolerance $\dot{a}(t)$:

$\dot{a}(t) = \eta\big(1-a(t)\big) - \mu a(t)\phi\big(z(t)\big)$, where $\eta > 0$ denotes the recovery rate of tolerance when pressure is low, and $\mu > 0$ denotes the depletion intensity caused by pressure. Equation (1) in Ref. [6] is a multiplicative equation for resident population $\boldsymbol{R}$, with resident population as the state; this paper reformulates it as a normalized social tolerance a, where $\eta(1-a)$ represents relaxation toward the good state after pressure is removed, and $\mu a\phi(z)$ represents exposure-induced depletion, retaining the core nonlinear feedback with a single state. Equation (8) has only one state variable, yet retains the most critical nonlinear feedback in Ref. [6].

Proof: When $a = 0$, $\dot{a} = \eta > 0$; when $a = 1$, $\dot{a} = -\mu\phi(z) \leq 0$. Hence, the solution always remains within [0,1]. Treating z as a given input, $f(a) = \eta(1-a) - \mu a\phi(z)$ is strictly decreasing on [0,1] because $f'(a) = -\eta - \mu\phi(z) < 0$). Therefore, there is a unique root $a^*(z) = \frac{\eta}{\eta+\mu\phi(z)} \in [0,1]$, and f is positive on $[0, a^*)$ and negative on $[a^*, 1]$. That is, $a^*(z)$ is the globally attracting equilibrium point for this fixed z. Its closed-form solution is

$$a(t) = a^*(z) + \Big(a(0) - a^*(z)\Big) e^{-(\eta + \mu\phi(z))t}$$

where the decay rate $\eta+\mu\phi(z)$ is constant, and $e^{-(\eta+\mu\phi(z))t}$ is the standard exponential convergence; overall, the solution converges globally and exponentially to $a^*(z) \in [0,1]$.

From Eq. (9), $a^{n+1} = a^n + \Delta t[\eta(1-a^n) - \mu a^n\phi(z^n)] = \big[1-\Delta t\big(\eta+\mu\phi(z^n)\big)\big]a^n + \Delta t\eta$. Since $0 \leq \phi \leq 1$, when $\Delta t(\eta+\mu) \leq 1$, the coefficient $1-\Delta t(\eta+\mu\phi) \geq 0$ is nonnegative, and the right-hand side is an increasing affine function of a^n. Substituting $a^n = 0$ gives $a^{n+1} = \Delta t\eta \geq 0$, and substituting $a^n = 1$ gives $a^{n+1} = 1 - \Delta t\mu\phi(z^n) \leq 1$. By induction, $a^n \in [0,1]$. This completes the proof.

Appendix B: Twenty-Edge Network Parameters and Summation of the Critical Threshold

Table 4 lists the geometric and social parameters for all 20 directed edges in the network shown in Figure 1. The free-flow traversal time is $T_e = L_e / v_f$ ($v_f = 0.9m/s$, converted from seconds to minutes); $b_e = (\mathbf{B}\pi)_e$ is the proportion of all tourists who pass through edge e, where $\mathbf{B}$ is the edge–path incidence matrix, and the fixed path proportions are taken as $\pi = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)^{\mathrm{T}}$. Each of the three themed routes is a closed loop (the edge sequences are given in the note below the table), so $b_e = \frac{1}{3}$ for every edge within a route. The cross-route connector edge e_{19} (Weitongxing Lane, $N_1 \to N_5$) does not belong to any fixed loop, so $e_{19} = 0$; it serves only as an alternative fork in dynamic Logit route choice. The sum of the term-by-term products in the last column is the critical exposure coefficient $H = \sum_e \omega_e b_e T_e$.

Table 4 Twenty-edge parameters and the summation table for $H = \sum_e \omega_e b_e T_e$

Route	Edge	From–To	Street/Alley Name	L_e (m)	w_e (m)	T_e (min)	ω_e	b_e	$\omega_e b_e T_e$ (min)
Red Spark Route	e_0	$N_0 \to N_1$	South Section of Buyegongsuo Lane	90	2.2	1.6667	1	0.333	0.5556
	e_1	$N_1 \to N_2$	North Section of Buyegongsuo Lane	60	2.2	1.1111	1	0.333	0.3704
	e_2	$N_2 \to N_3$	Buyegongsuo Lane Extension	50	2.2	0.9259	1	0.333	0.3086
	e_3	$N_3 \to N_4$	Boxian Park Passage	160	3.5	2.963	0.4	0.333	0.3951
	e_4	$N_4 \to N_0$	Peripheral Return	260	4	4.8148	0.1	0.333	0.1605
Charity Culture Rout	e_5	$N_0 \to N_5$	East Section of Banbi Lane	110	2.6	2.037	1	0.333	0.679
	e_6	$N_5 \to N_6$	West Section of Banbi Lane	140	2.6	2.5926	1	0.333	0.8642
	e_7	$N_6 \to N_7$	Sanshan Lane	120	2.9	2.2222	0.7	0.333	0.5185
	e_8	$N_7 \to N_8$	West Section of Huoxingmiao Lane	80	3.8	1.4815	0.7	0.333	0.3457
	e_9	$N_8 \to N_9$	Middle Section of Huoxingmiao Lane	70	3.8	1.2963	0.7	0.333	0.3025
	e_{10}	$N_9 \to N_{10}$	East Section of Huoxingmiao Lane	70	3.8	1.2963	0.7	0.333	0.3025
	e_{11}	$N_{10} \to N_0$	Huoxingmiao Lane → Xixiejiao	130	3.3	2.4074	1	0.333	0.8025
Silver-Age Life Route	e_{12}	$N_0 \to N_{10}$	Xiexiejiao → Wanjia Lane	130	3.3	2.4074	1	0.333	0.8025
	e_{13}	$N_{10} \to N_{11}$	North Section of Wanjia Lane	80	3.1	1.4815	1	0.333	0.4938
	e_{14}	$N_{11} \to N_{12}$	Middle Section of Wanjia Lane	100	3.1	1.8519	1	0.333	0.6173
	e_{15}	$N_{12} \to N_{13}$	Street and Alley Theater Link	70	2.5	1.2963	0.7	0.333	0.3025
	e_{16}	$N_{13} \to N_{14}$	Dalongwang Lane Living Block	90	3.7	1.6667	1	0.333	0.5556
	e_{17}	$N_{14} \to N_{15}$	Meal Assistance Point Link	60	2.4	1.1111	0.7	0.333	0.2593
	e_{18}	$N_{15} \to N_0$	Meal Assistance Point Return to Station	100	3	1.8519	0.7	0.333	0.4321
Contact Point	e_{19}	$N_1 \to N_5$	Weitongxing Lane Link	120	2	2.2222	0.7	0	0
			Total	$\sum_e b_e T_e = 12.1605$		—	—		$H = 9.0679$

Note: The three fixed loop routes are Route 1 (Red Spark), $e_0 \to e_1 \to e_2 \to e_3 \to e_4$; Route 2 (Charity and Culture), $e_5 \to \ldots \to e_{11}$; and Route 3 (Silver-Age Living), $e_{12} \to \ldots \to e_{18}$. Their free-flow travel times are 11.5, 13.3, and 11.7 min, respectively. Thus, $\sum_e b_e T_e = \frac{11.5+13.3+11.7}{3} = 12.1605$ min. After incorporating the sensitivity weights, $H = \sum_e \omega_e b_e T_e = 9.0679$ min, and the average weight reduction is $\frac{9.0679}{12.1605} = 0.746$. Setting the steady-state tolerance exactly at its floor, $a^* = a_{min} = 0.6$, and back-solving Eq. (11) gives the critical exposure $z^* = 0.0390$. Substituting this into Eq. (12) gives $\lambda^* = \frac{R_{res} z^*}{H} = \frac{750 \times 0.0390}{9.0679} = 3.23$ 人/minpersons/min, consistent with Section 8.2 of the main text. e_{11} ($N_{10} \to N_0$) and e_{12} ($N_0 \to N_{10}$) are opposite-direction edges of the same physical street, and the receiving function in Eq. (3) subtracts the opposite-direction flow $N_{\bar{e}}$ over this pair.